

TRANSFER CAPABILITIES of BLENDGAN by LEVERAGING INTERFACEGAN

Ali Osman Berk Şapcı - 21022

Cem Kaya - 26440

Furkan Reha Tutaş - 21036

TABLE OF CONTENTS

ABSTRACT	2
LITERATURE REVIEW	2
Generative Adversarial Network (GAN)	2
BlendGAN	2
InterFaceGAN	3
PROBLEM DEFINITION	3
ENCODING for MANIPULATION in to the LATENT SPACE of INTERFACEGAN	4
Approach 1	5
Approach 2	5
Approach 3	6
RESULTS	7
CONCLUSION	10
REFERENCES	10

ABSTRACT

In recent years, there has been a significant improvement in Generative Adversarial Networks (GANs). Although many papers propose different solutions, this paper focuses on InterFaceGAN and BlendGAN. InterFaceGAN proposes semantic face editing by interpreting the latent semantics learned by GANs. On the other hand, BlendGAN proposes arbitrary stylized face generation by leveraging a flexible blending strategy and a generic artistic dataset. This paper proposes semantically editing a given image using the latent space of a GAN and then stylizing the edited image for a given art style. To achieve this, InterFaceGAN was used for semantic face editing whereas BlendGAN was used for stylizing the edited image given a reference comic style. Finally, the paper analyzes the attribute transfer capabilities of the BlendGAN network by leveraging InterFaceGAN.

LITERATURE REVIEW

Generative Adversarial Network (GAN)

In recent years, there have been increasing attention & advanced improvements in GAN since it is good for generating high-quality photo-realistic images. GAN relies on adversarial training between generator and discriminator which also is the key to its success.

BlendGAN

BlendGAN is a framework that can generate stylized face images with high quality. The framework can be summarized as follows:

1. The style encoder learns the style code of the given style image
2. The style code is normally distributed according to the gaussian distribution
3. Two MLPs (Multi-layer Perceptrons) transform face & style codes into separate spaces.
4. Separate Spaces are combined by a weighted blending module (WBM)
5. WBM fed into generator to produce natural & stylized face images
6. There are three discriminators used to distinguish whether the image is real or fake;
 - a. Face Discriminator: Real vs Fake Natural-Face
 - b. Style Discriminator: Real vs Fake Stylized-Face
 - c. Style_Latent Discriminator: Stylized-Face Image vs Style Latent Code

InterFaceGAN

InterFaceGAN is an approach to turn unconditionally trained face synthesis models to controllable GANs. The controllability enables semantic face editing. This is achieved by:

1. Taking a pretrained GAN model and semantic attribute scorer
2. Generating faces and scoring the desired attributes for sampled latent vectors
3. Training a linear SVM on the input latent vectors and corresponding attribute scores
4. Using the normal vector of the attribute boundary hyperplane to edit an image by doing vector arithmetic on the input images latent vector
 - a. There are numerous ways to encode an image to the editable latent space of a GAN and since the most performant GAN in the InterFaceGAN pretrained boundaries is StyleGAN there are multiple latent spaces that an image can be encoded in.

PROBLEM DEFINITION

The problem is to semantically edit a given image using the latent space of a GAN and then stylize the edited image for a given art style. The aim is to analyze the attribute transfer capabilities of the used BlendGAN network by leveraging InterFaceGAN.

PROPOSED APPROACH

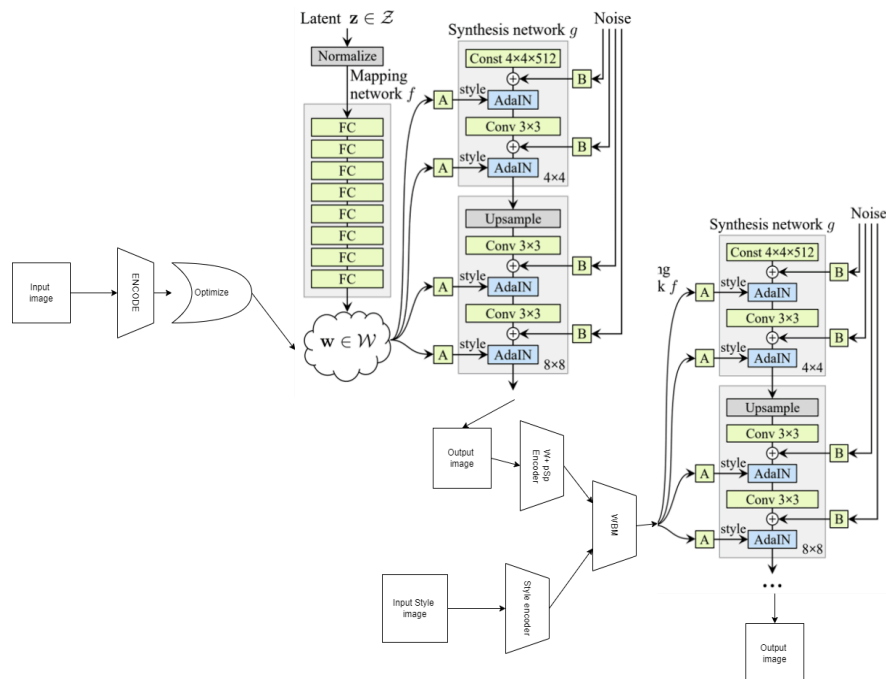


Figure 1: Flowchart of Proposed Approach

For semantic face editing, InterFaceGAN is used. It achieves face editing by interpreting the latent semantics learned by GANs. It can manipulate gender, age, expression, the presence of eyeglasses, and face poses. InterFaceGAN can also be used to fix artifacts of images generated by GANs.

BlendGAN is used for stylizing the edited image given a reference comic style. BlendGAN can be used for arbitrary stylized face generation by leveraging a flexible blending strategy and a generic artistic dataset. Another aim is to benefit from a novel large-scale artistic face dataset AAHQ, which is proposed by Liu, Mingcong, et al.

ENCODING for MANIPULATION in to the LATENT SPACE of INTERFACEGAN

InterFaceGAN semantic image manipulation results show that it can edit a sampled image to achieve desired effects for trained boundaries. However, it relies on external Gan inversion methods. As discussed in "GAN Inversion: A Survey" there are 3 main ways to encode an image for a given latent space:

1. Training an encoder:

- a. Sample latent vectors and train an encoder to predict this latent vector for the corresponding image.

- b. pSp (pixel2style2pixel) training an encoder unsupervised. This is achieved by using the generator of the GAN as the decoder of an autoencoder and the latent vector as the bottleneck layer and training on the GANs domain data.

2. Optimizing a latent vector: Start with a random latent vector and optimize it by backpropagating through the GAN

3. Hybrid: instead of starting with a random vector this approach starts with an encoder predicted latent vector and optimizes it.

Encoding Results

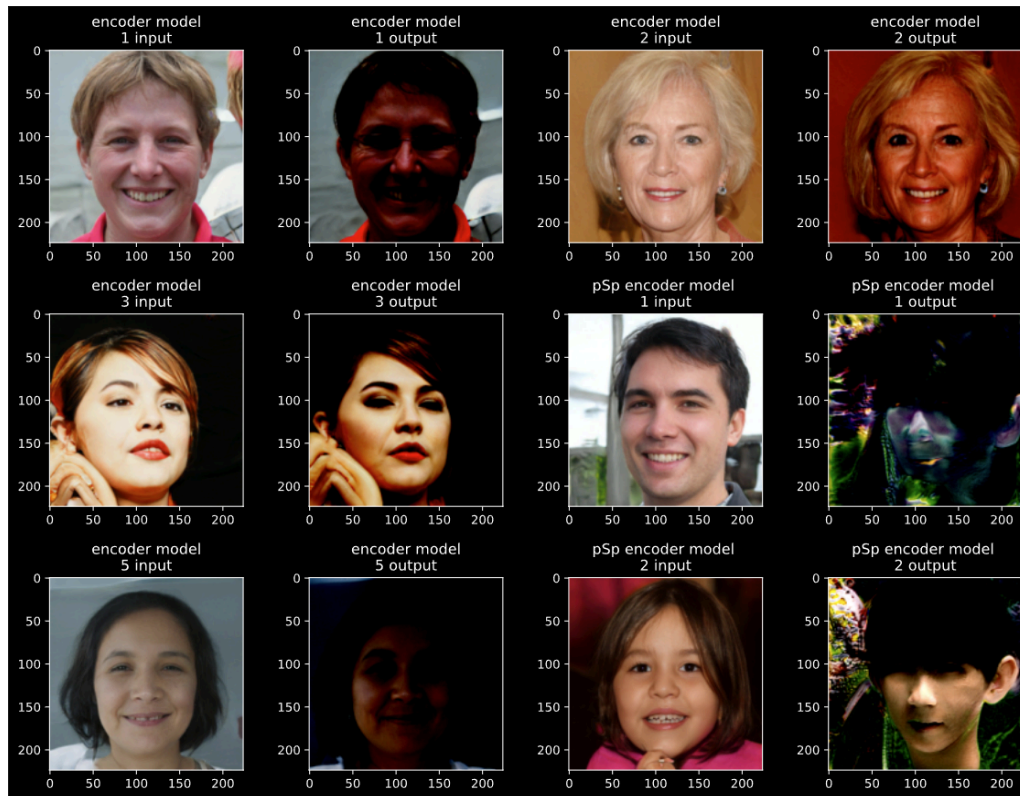


Figure 2: Trained Encoder Results

Approach 1

The nan-pSp encodings above were attained by a resnet 38 which was trained on 50k image w- space latent vector (1,512) pairs. The images can be semantically similar however they are not photorealistic nor the same person. The pSp encoders were trained on the commercially available portion of FFHQ (52K images). However, due to the computational cost of training such an encoder they failed to converge. (3.2h per epoch on a gpu with max batch size = 4 on 8GB).

Approach 2

Optimizing a random latent vector by backpropagating through the GAN. For this we chose w+ (18,512) latent space since "GAN Inversion: A Survey " shows that it is possible to get high fidelity reconstructions using this space. However, as they mentioned without a close enough starting point the outputs are not photorealistic. This approach takes 5 min per image.

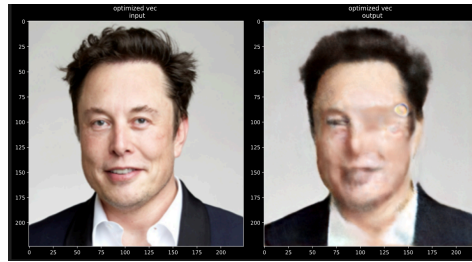


Figure 3: Sample Result for Approach 2

Approach 3

Hybrid encoding encoder + optimization. Using an encoder which predicts a W space encoding and then optimizing this encoding. This approach produced photo realistic and semantically similar outputs however even though the outputs were semantically close they were not the same person.

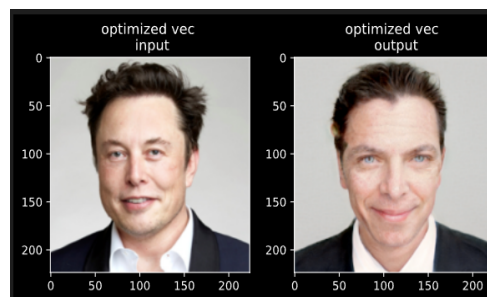


Figure 4: A middle aged smiling man with blue eyes, short hair and a black suit on.



Figure 5: Evolution of optimized image.

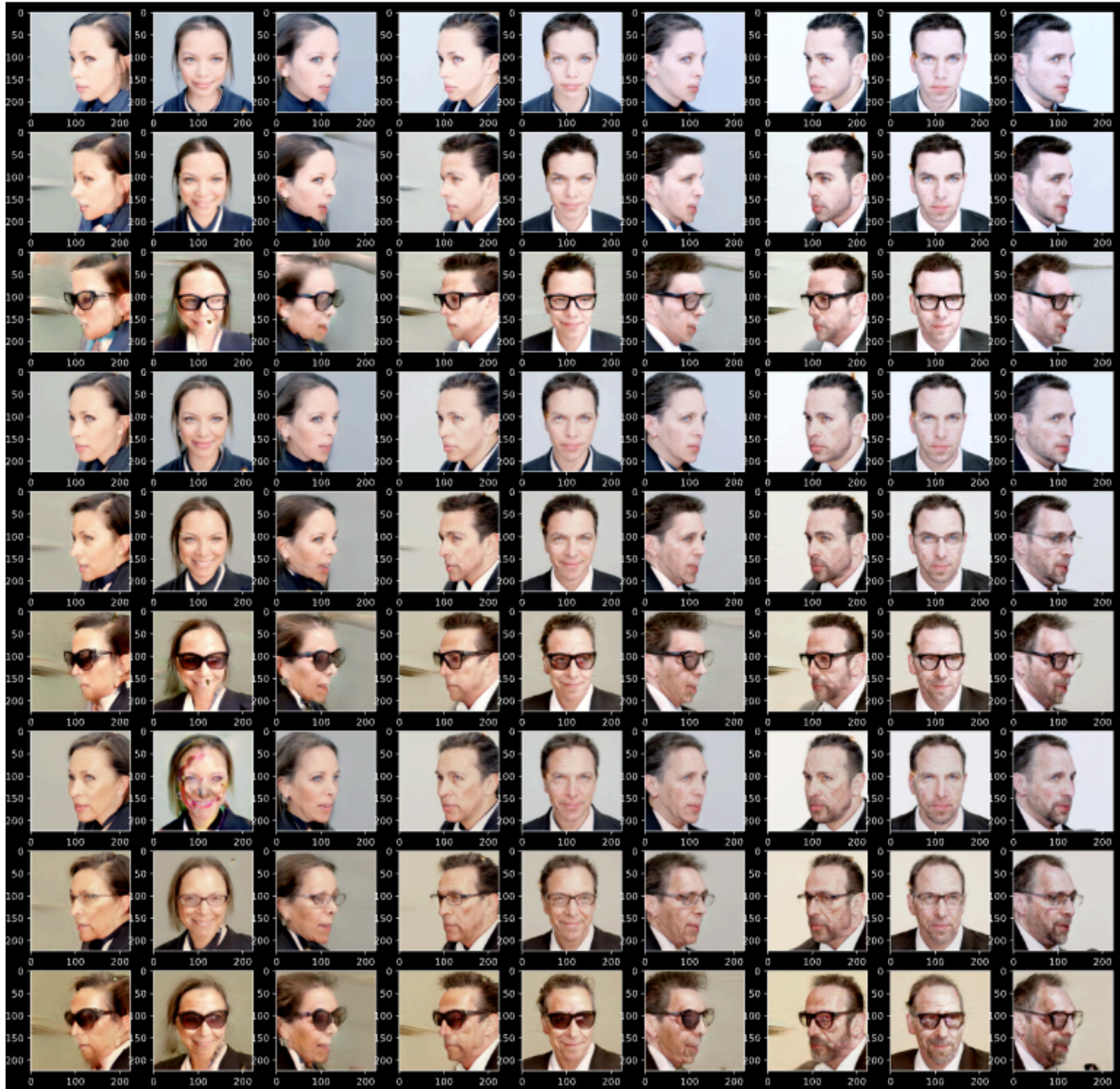


Figure 5: Sample result for Approach 3

These latent vectors were stable under semantic editing. Which is not a given for every gan inversion method.

RESULTS

We started by exploring the InterfaceGAN's ability of attribute editing and its limitations. The model allows modifying five different attributes, namely age, eyeglass, gender, smile and pose, i.e., head position. In the animations in Figure 2., it can be seen that InterfaceGAN is able to robustly edit different attributes with respect to a given corresponding weight. It starts to introduce artifacts when the weights go to minus infinity or infinity, as expected.

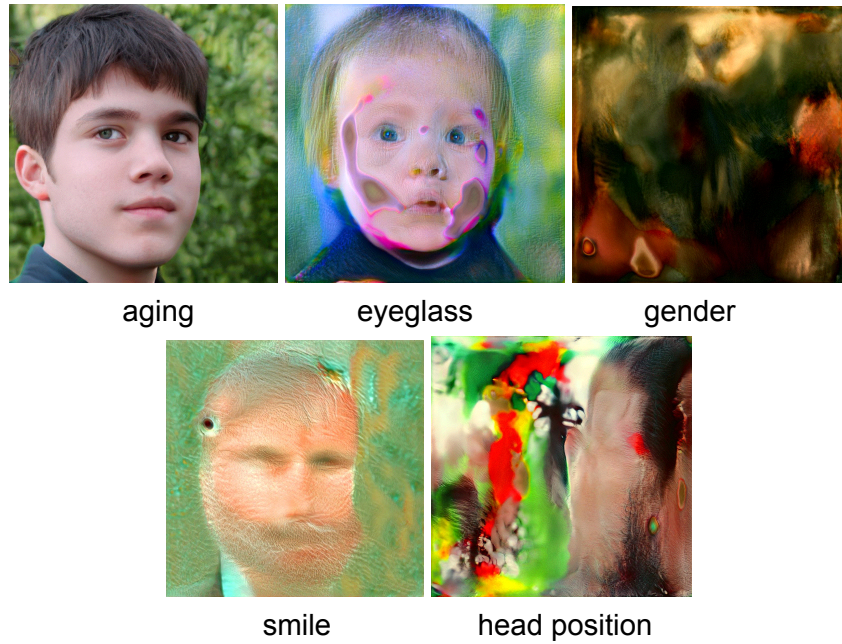


Figure 6: Animations demonstrating attribute editing with varying degrees of InterfaceGAN.

After sampling portraits without any modified attributes, we generated images with different edited attribute combinations. We used the same weight, and its negative counterpart, for each attribute. In other words, we set age weight either 2, or -2, and similarly for others, in order to keep the generated image set finite.

Then, using edited images, we “styled” them with different reference styles using BlendGAN. A number of different styles are used in this procedure, some of them chosen from the test images given with BlendGAN (from artistic face dataset *AAHQ*). We also collected some style references from various comics such as *Hellboy* and *Watchmen*. BlendGAN successfully handled almost all tested styles from *AAHQ*, but failed to blend the style of some comic images. This is most probably due to the background of those style images and complex textures.

In Figure 3, we demonstrated styled portraits of an adult woman, with many different edited attribute combinations. For ease of comparison, in the Figures 3a, 3b, and 3c, we used a single reference style, a GIF of styled portraits with multiple edited attributes and multiple references is given in Figure 3d.



(a) Original sampled image, no attribute is edited.



(b) Younger version and corresponding styled portrait.



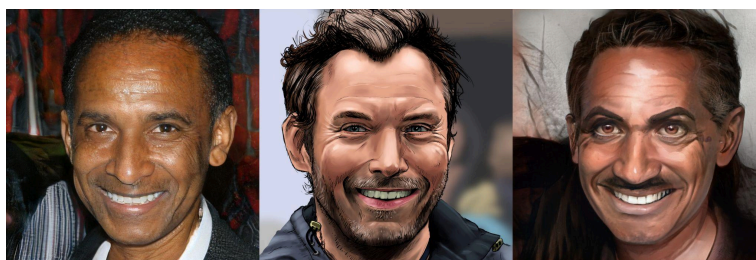
(c) A version with eyeglass and corresponding styled portrait.



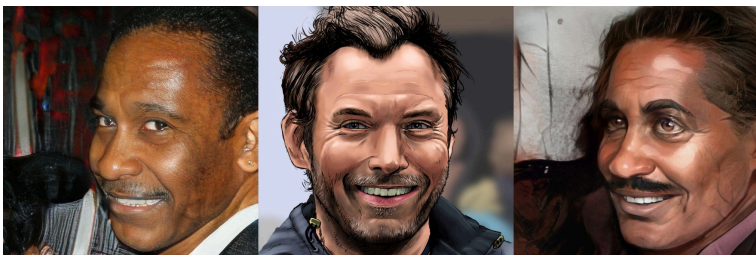
(d) Older version with eyeglass and corresponding styled portraits with multiple reference styles.

Figure 7: Styled portraits of an adult woman, with different edited attributes.

In Figure 4, portraits of an adult with different orientations is given with a comic-like style reference. Both InterfaceGAN and BlendGAN successfully generate corresponding versions.



(a) Original sampled portrait of an adult man, orientation is straight towards the camera and corresponding comic-like styled image.



(b) Edited portrait, orientation of the person is towards the left of the camera, and corresponding comic-like styled image.



(c) Edited portrait, orientation of the person is towards the right of the camera, and corresponding comic-like styled image.

Figure 8: Styled portraits of an adult man, demonstrating the pose editing abilities of InterfaceGAN.

CONCLUSION

GANs are useful for image manipulation and stylization. As experienced in this piece of work, to realize real life use cases better encoding techniques may be needed. There are 3 different approaches used to improve the performance of GAN. Approach 3 was the best performer but it requires 5 minutes per an image and it uses a psp encoder technique which is hard to train.

REFERENCES

- InterfaceGAN Repository: <https://github.com/genforce/interfacegan>
- InterfaceGAN Paper (TPAMI): <https://arxiv.org/pdf/2005.09635.pdf>
- InterfaceGAN Paper (CVPR): <https://arxiv.org/pdf/1907.10786.pdf>
- GAN Inversion: A Survey: <https://arxiv.org/abs/2202.07145>
- BlendGAN Repository: <https://github.com/onion-liu/BlendGAN>
- BlendGAN Paper (NIPS): <https://arxiv.org/abs/2110.11728>

Liu, Mingcong, et al. "BlendGAN: Implicitly GAN Blending for Arbitrary Stylized Face Generation." *Advances in Neural Information Processing Systems* 34 (2021).

Shen, Yujun, et al. "Interpreting the latent space of gans for semantic face editing." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2020.